

## A. Structure-derived contact pretraining

PDB RNA-ligand  
complexes



Distance labels  
< 4 Å contacts



Contact-supervised  
DeepRSMA pretraining



Contact-pretrained  
backbone



## B. Transfer to affinity prediction

R-SIM affinity data  
RNA + SMILES + pKd



pKd fine-tuning  
MSE loss



Affinity prediction  
PCC / SCC / RMSE